Bilal Aytekin

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RESEARCH INTERESTS

Number theory, arithmetic geometry, and formalization of mathematics; in particular, motivic height zeta functions and weighted projective stacks.

EDUCATION

University of Maryland, College Park — 2023 – present
Ph.D. in Mathematics — College Park, MD
- Advisor: Dori Bejleri.

Boğaziçi University — 2018 – 2023
B.Sc. in Mathematics and B.Sc. in Computer Engineering (double major) — Istanbul, Turkey
- Thesis (Computer Engineering): *Entropy Hierarchy*. Advisor: Cem Say.

PUBLICATIONS AND PREPRINTS

1. **Higher-genus motivic height zeta functions for weighted projective stacks**.
   In preparation.
2. **MathAdv: What Theorem Provers Know, Reason, Formalize, and Generalize**.
   With J. Yuan, C. Martinez Lockhart, X. Liu, J. Wang, C. Deng, X. Han, V. Mastrantonis, D. Gudin, S. Zhu, A. Mohamed, J. Lang, Z. Song, and F. Huang. Submitted, 2026. arXiv:2608.25449.
3. **An Alexandrov Topology for Maximal Cohen–Macaulay Modules**.
   With M. Akdenizli, B. Çetin, and Ö. Esentepe. *Journal of Pure and Applied Algebra* 228 (2024), no. 1. arXiv:2210.03532.

TALKS

First Chern Classes — Mar 2026
Student 6-Functor Formalism Seminar, University of Maryland

From Quadratic Functions to Modular Functions — Dec 2025
Student Algebra and Number Theory Seminar, University of Maryland

Local and Global Root Numbers of Elliptic Curves — Jun 2022
Summer School on Applied Arithmetic

Torsion Subgroups of Elliptic Curves — Sep 2021
Directed Reading Program Turkey Symposium

TEACHING

University of Maryland — 2023 – present
Department of Mathematics — College Park, MD

Instructor: Introduction to Math Modeling and Probability	Fall 2025
Teaching Assistant: Linear Algebra	Spring 2026
Teaching Assistant: Differential Equations	Spring 2025
Teaching Assistant: Calculus II	Fall 2024
Teaching Assistant: Calculus III	Spring 2024
Teaching Assistant: Calculus II	Fall 2023

Boğaziçi University — 2022
Departments of Mathematics and Computer Engineering — Istanbul, Turkey

Undergraduate Assistant: Graduate Algebra II	Fall 2022
Grader: Introduction to Probability and Statistics for Computer Engineers	Fall 2022

CONFERENCES AND WORKSHOPS

JAVA 2025, 2026
Algebraic Geometry NorthEastern Series (AGNES)
Algebraic Geometry Bootcamp
Arizona Winter School

AWARDS AND HONORS

TÜBİTAK 2205 Undergraduate Scholarship 2019 – 2023
Scientific and Technological Research Council of Turkey (*TÜBİTAK*)
Awarded to undergraduate science students who ranked highly in the national university entrance exam.

COMPUTATIONAL SKILLS

Programming: Python, C++, Java
Formal verification: Lean 4, Mathlib
Computer algebra: SageMath, Magma, PARI/GP
Typesetting: LaTeX

Last updated November 2026.